Full Stack Assignment – 32

Assignment ID: FSD-32

Title: Internal Training Enrollment Management System

Duration: 3 Hours

1️⃣ Objective

Design and develop an **Internal Training Enrollment Management System** that allows employees to enroll in training programs and trainers to manage enrollments.
The system must enforce seat limits and ensure **data persistence using a database**.

2️⃣ User Roles

EMPLOYEE

- Register and log in to the system
- View available training programs
- Enroll in training programs
- View enrollment status

TRAINER

- Register and log in to the system
- Create and manage training programs
- View enrolled employees

3️⃣ Functional Requirements

Authentication & Authorization

- Implement user registration and login
- Use **JWT-based authentication**
- Protect all backend APIs using JWT
- Enforce **role-based access control** on APIs and frontend views

Training Management

- TRAINER can create training programs with seat limits
- Training data must be stored in the database
- Training availability must be fetched dynamically

Training Enrollment

- EMPLOYEE can enroll in available training programs
- Enrollment data must be stored in the database

- EMPLOYEE can view enrollment status

4️⃣ Business Rule (Critical)

⚠️ **Enrollment must be rejected once the seat limit is reached.**

No additional enrollments must be allowed beyond the defined capacity.

5️⃣ Technical Constraints (Mandatory)

- All training and enrollment data **must persist in a database**
- Hardcoded or in-memory storage is **not allowed**
- Backend APIs must reflect actual database state
- JWT authentication and role validation are mandatory
- Use environment variables for sensitive configuration

6️⃣ Frontend Requirements

- Login and registration pages
- Role-based dashboards for EMPLOYEE and TRAINER
- Training listing and enrollment view for EMPLOYEE
- Training creation and enrollment overview for TRAINER

UI design is not evaluated; rule enforcement and correctness matter.

7️⃣ Submission Requirements (Strict)

GitHub Repository

- Backend source code
- Frontend source code
- Proper folder structure
- Clear and meaningful commit history

Live Deployment

- Live frontend deployment link
- Live backend deployment link
- Backend must be connected to the database

README.md (Mandatory)

The README file must include:

- Project overview
- Tech stack used
- User roles and permissions

- API endpoints
- Database schema or collections
- Steps to run the project locally
- Live frontend and backend links

8 Evaluation Note

Evaluation focuses on **capacity enforcement, role control, and end-to-end execution**.